

Stat 532 Final Project

Survival Analysis of Heart Transplant Data

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1 Abstract

The Stanford Heart Transplant data contains the survival of patients on the waiting list of the Stanford heart transplant program. Here, we analyze the survival of 103 patients on the waiting list for the Stanford heart transplant program. The purpose of this report is to outline the results of the survival analysis on the heart transplant data. Analyses are done using Kaplan-Meier estimates and fitting a Cox-proportional hazards model. The Kaplan-Meier estimates show the survival time of individuals who have received prior bypass surgery are higher in comparison to those who have not had a history of bypass surgery. Additionally, using the Kaplan-Meier estimate we see that those who have received a heart transplant have a higher survival time compared to individuals who have not. Furthermore, from the Cox-proportional hazards model, we conclude that the transplant variable is indeed time-dependent, and there are interactions between factors in the data, specifically between year and transplant, which must be included into the Cox-proportional hazard model.

2 Introduction

Heart transplants are typically performed when other treatments for heart problems have not shown improvement in health. In adults, heart failure can be caused by many reasons, including if you have some weakening of the heart muscle, (cardiomyopathy), or if you are born with a heart problem (congenital heart defect). In children, heart failure is often caused by cardiomyopathy and congenital heart defects. It is important to note that many other factors are considered when being put on the waitlist to receive a heart transplant in real life (1). The Stanford Heart Transplant data is a dataset detailing information of patients on the waiting list for the Stanford heart transplant program, including statistics on survival after acceptance into the transplant program.

The so-called Kaplan-Meier estimate, also known as the "product-limit estimator", is one of the simplest ways of calculating the survival of individuals across time (2). This model also describes the survival of patients between given populations, in our case, being within the population of individuals who have received prior bypass surgery against the population of individuals who have not received prior bypass surgery. As well, the survival of patients between the transplant groups can be compared using the Kaplan-Meier estimates. This is a very basic approach, which we use to look at a simple difference in survival between the bypass surgery groups and between the transplant groups.

For this report we also be look at the Cox-proportional hazard model. This model is a regression-based model that is commonly used to analyze the association between survival of individuals and predictor variables across time (3). Furthermore, when some covariates are found to be time-dependent based on diagnostic plots, they will be adjusted accordingly in our analysis. Additionally, after considering the biological interpretations of our predictor variables, interactions are added into the final model.

In this research, the Stanford heart transplant data is used to fit two Kaplan-Meier estimates of different population groups and to fit the aforementioned Cox-proportional hazard model. The objective of this study is to determine the survival probability of individuals who were placed on the heart transplant list for the Stanford heart transplant program.

3 Methods

3.1 Description of Data

The heart dataset is found within the `survival` package in R. For this report, the `jaso` and `heart` data will be used. The original dataset `jaso` contains the observations in the paper from 1977 by J. Crowley and M. Hu, where they conducted a covariance analysis using this data (4). The data is left truncated and right censored. For the Kaplan-Meier estimates, `jaso` data was used. The `heart` contains the pre-processed data of `jaso`, which was used to build the Cox-proportional hazard models.

Summary of the `jaso` data in figure 1.

birth.dt	accept.dt	tx.date	fu.date	fustat	surgery	age
Min. :1905-04-02	Min. :1967-09-13	Min. :1968-01-06	Min. :1967-09-18	Min. :0.0000	Min. :0.0000	Min. : 8.786
1st Qu.:1919-06-01	1st Qu.:1969-04-29	1st Qu.:1969-09-14	1st Qu.:1970-01-26	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:41.180
Median :1924-03-30	Median :1971-03-24	Median :1971-10-13	Median :1972-05-22	Median :1.0000	Median :0.0000	Median :47.751
Mean :1925-12-07	Mean :1971-02-07	Mean :1971-06-14	Mean :1971-12-13	Mean :0.7282	Mean :0.1553	Mean :45.169
3rd Qu.:1930-03-19	3rd Qu.:1972-10-01	3rd Qu.:1973-01-17	3rd Qu.:1974-03-25	3rd Qu.:1.0000	3rd Qu.:0.0000	3rd Qu.:52.059
Max. :1960-07-21	Max. :1974-03-22	Max. :1974-03-31	Max. :1974-04-01	Max. :1.0000	Max. :1.0000	Max. :64.408
NA's :34						
futime	wait.time	transplant	mismatch	hla.a2	mscore	reject
Min. : 0.0	Min. : 0.00	Min. :0.0000	Min. :1.000	Min. :0.0000	Min. :0.000	Min. :0.0000
1st Qu.: 32.5	1st Qu.: 9.00	1st Qu.:0.0000	1st Qu.:2.000	1st Qu.:0.0000	1st Qu.:0.750	1st Qu.:0.0000
Median : 89.0	Median : 25.00	Median :1.0000	Median :3.000	Median :0.0000	Median :1.080	Median :0.0000
Mean : 309.2	Mean : 37.68	Mean :0.6699	Mean :2.708	Mean :0.2615	Mean :1.165	Mean :0.4461
3rd Qu.: 411.5	3rd Qu.: 45.00	3rd Qu.:1.0000	3rd Qu.:3.000	3rd Qu.:1.0000	3rd Qu.:1.580	3rd Qu.:1.0000
Max. :1799.0	Max. :309.00	Max. :1.0000	Max. :4.000	Max. :1.0000	Max. :3.050	Max. :1.0000
NA's :34						
NA's :38						
NA's :38						
NA's :38						
NA's :38						

Figure 1: Summary of Variables in `jaso`. See appendix for definitions.

The `heart` from the Stanford heart transplant from the 1977 study contains information on 103 patients who are placed on the Stanford heart transplant list. This dataset includes the following factors: start, stop, event, age, year, surgery, transplant, and id. These variables are summarized in figure 2.

start	stop	event	age	year	surgery	transplant	id
Min. : 0.00	Min. : 1.0	Min. :0.000	Min. : -39.2142	Min. : -0.04928	Min. :0.0000	0:103	Min. : 1.00
1st Qu.: 0.00	1st Qu.: 17.0	1st Qu.:0.000	1st Qu.: -6.6468	1st Qu.: 1.66051	1st Qu.:0.0000	1: 69	1st Qu.: 28.00
Median : 0.00	Median : 45.5	Median :0.000	Median : -0.1136	Median : 3.75086	Median :0.0000		Median : 53.50
Mean : 15.51	Mean : 201.3	Mean :0.436	Mean : -2.4840	Mean : 3.45329	Mean :0.1686		Mean : 53.06
3rd Qu.: 18.25	3rd Qu.: 181.5	3rd Qu.:1.000	3rd Qu.: 4.0459	3rd Qu.: 5.03491	3rd Qu.:0.0000		3rd Qu.: 78.25
Max. :310.00	Max. :1800.0	Max. :1.000	Max. : 16.4079	Max. : 6.47228	Max. :1.0000		Max. :103.00

Figure 2: Summary of Variables in `heart`.

Definitions of the variables in `heart` are summarized in the table below:

Table 1: Definition of Variables in `heart` dataset.

Variables	Definitions
start, stop, event	Entry and Exit time and status for this interval of time
age	age - 48 years
year	year of acceptance (in years after Nov 1 1967)
surgery	prior bypass surgery (1 = Yes, 0 = No)
transplant	received transplant (1 = Yes, 0 = No)
id	patient id height

The heart data is formatted such that it contains information of patients who have received a transplant surgery in two separate rows. An example of patient 3 is shown in figure 3.

	start	stop	event	age	year	surgery	transplant	id
3	0.0	1.0	0	6.29705681	0.26557153	0	0	3
4	1.0	16.0	1	6.29705681	0.26557153	0	1	3

Figure 3: Information of patient 3 in heart.

Here, it is shown that the patient waited 1 day on the transplant list, received a transplant, passed away 15 days later.

3.2 Kaplan-Meier Estimates

Note that the equations and descriptions are taken from references (5) and (6).

As noted in the introduction, the Kaplan-Meier estimate is a simple method to describe the survival of individuals over time.

To calculate this estimate, the following variables are defined (same definitions as in (6)):

- t_i : Event times $t_1 < \dots < t_D$. Here, D is distinct times.
- d_i : Number of events (deaths) at time t_i
- Y_i : Number of individuals who are alive/experience the event at time t_i
- $\frac{d_i}{Y_i}$: Estimate of conditional probability that an individual who survives just prior to time t_i experiences the event at time t_i

The Kaplan-Meier estimator is a step function that provides an estimate of the survival function. It is calculated using the following equation:

$$\hat{S}(t) = \prod_{t_i \leq t} (1 - \frac{d_i}{Y_i}) \quad (1)$$

with corresponding variance given by Greenwood's formula:

$$\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_i \leq t} \frac{d_i}{Y_i(Y_i - d_i)} . \quad (2)$$

The standard error of the Kaplan-Meier estimate is thus given by the square root of equation 2.

The Kaplan-Meier estimate of the both transplant and surgery groups were calculated with corresponding curves plotted (7).¹ To examine the survival as a function of whether the patient received a heart transplant, or had a history of prior bypass surgery, a log-rank test was conducted.

A log-rank test has the following hypothesis:²

- H_0 : There are no differences in the Kaplan-Meier estimates between groups.
- H_A : There are differences in the Kaplan-Meier estimates between groups.

¹Kaplan-Meier estimates were calculated using the `surfdiff` function (8) with corresponding curves plotted via `ggsurvplot`

²The p-value of the log-rank test was calculated by indicating `pval = T` with `ggsurvplot`.

3.3 Cox-Proportional Hazards Model

Again, note that the equations and descriptions are taken from references (5) and (6).

We also build a Cox-proportional hazard model to assess the ratio of hazard rates across time for two or more individual groups.³

To calculate this estimate, the following terms must be noted (same definitions as in (6)):

Let X denote time to event. The data consists of the triple $(T_j, \delta_j, Z_j(t)), j = 1, \dots, n$ where

- T_j : Time on study for j th patient.
- δ_j : Indicator for event (death) for j th patient
- $Z_j(t)$: $(Z_{j1}(t), \dots, Z_{jp}(t))^T$ is the vector of parameters or risk factors for the j th individual at time t which may affect the survival distribution of X . This vector may consist of both fixed and time-dependent covariates.

Additionally, the following is defined:

Let $h(t|Z)$ be the hazard rate at time t for a patient with risk parameter vector Z .

$$h(t|Z) = h_0(t)c(\beta^T Z) \quad (3)$$

where $h_0(t)$ is the baseline hazard rate, $\beta = (\beta_1, \dots, \beta_p)^T$ is the parameter vector, and $c(\beta^T Z) = \exp(\sum_{k=1}^p \beta_k Z_k)$.

Looking at two individuals with covariate vector Z and Z^* , the relative risk is free of time:

$$\frac{h(t|Z)}{h(t|Z^*)} = \frac{h_0(t)\exp(\sum_{k=1}^p \beta_k Z_k)}{h_0(t)\exp(\sum_{k=1}^p \beta_k Z_k^*)} = \exp[\sum_{k=1}^p \beta_k (Z_k - Z_k^*)] \quad (4)$$

Here, the relative risk is defined as the hazard ratio comparison between an individual with risk factor Z and another individual with risk factor Z^* .

In the heart data, the variables surgery and transplant will be coded as the following:

- Surgery

$h(t|Z = 1)$ for receiving prior bypass surgery

$h(t|Z = 0)$ for not receiving prior bypass surgery

- Transplant

$h(t|Z = 1)$ for receiving transplant

$h(t|Z = 0)$ for not receiving transplant

After the Cox-proportional hazard models are fit, the time-independent assumption is checked by plotting the Schoenfeld residuals.⁴

When there are indications of violations of the time assumption, models are adjusted accordingly with justifications.

³The Cox-proportional hazard models are fitted using the `coxph` function (8).

⁴We use `cox.zph` to plot the residuals.

3.4 Program

Everything was programmed in R. Please contact author at sp8@ualberta.ca for full coding script of this project.

4 Results

4.1 Kaplan-Meier Estimates

The Kaplan-Meier estimates were calculated to determine the differences in survival of individuals in different transplant and surgery groups. The results are shown below:

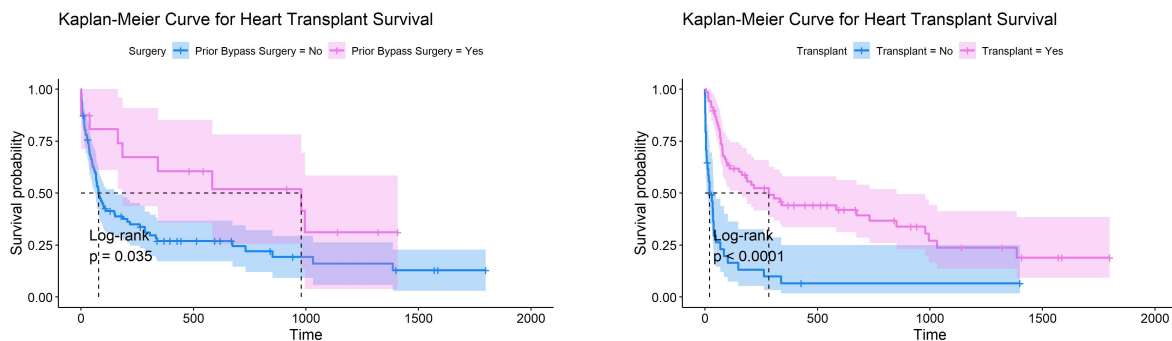


Figure 4: Plot of Kaplan-Meier estimates with labelled median survival time.

Left: Surgery group, Right: Transplant group

The dotted black lines indicate a median survival time for all groups. The colored bands imply 95% confidence intervals.

Between the individuals in respective surgery groups, it was shown that those who have received a prior bypass surgery indeed have a higher survival probability. This was also supported by the p-value of our log-rank test, $p < 0.035$, therefore at the 5% significance level, we rejected the null hypothesis that there were no differences in the Kaplan-Meier estimates between the groups, and concluded that there were differences between the groups. Furthermore, between the individuals in respective heart transplant groups, it was evident that those who have received a heart transplant indeed have a higher survival probability. This was supported by the p-value of the log-rank test, $p < 0.0001$. At the 5% significance level, we concluded similarly that there were differences between the groups.

4.2 Cox-Proportional Hazards Model

There are two Cox-proportional hazard models fitted for this project.⁵ Quoting directly from reference (8), Terry Therneau, Cynthia Crowson, and Elizabeth Atkinson describe time-dependent covariates in a Cox-proportional hazard model in the following way: “The key rule for time dependent covariates in a Cox model is simple and essentially the same as that for gambling: you

⁵The `cox.zph` function was used to test the proportionality assumption of all the predictors in the models using a log transformation.

cannot look into the future. A covariate may change in any way based on past data or outcomes, but it may not reach forward in time.”

First model, which turned out to be a very naive approach, considered all variables in the heart dataset. As seen in the p-values, age and year were found to be significant. At the 5% significance level, we rejected the null hypotheses of the likelihood ratio, Wald, and Score (log-rank) tests. Model summary is in figure 5.

```
call:
coxph(formula = surv(start, stop, event) ~ age + year + surgery +
      transplant, data = heart)

n= 172, number of events= 75
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
age	0.02717	1.02754	0.01371	1.981	0.0476 *
year	-0.14635	0.86386	0.07047	-2.077	0.0378 *
surgery1	-0.63721	0.52877	0.36723	-1.735	0.0827 .
transplant1	-0.01025	0.98980	0.31375	-0.033	0.9739

 signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age	1.0275	0.9732	1.0003	1.0555
year	0.8639	1.1576	0.7524	0.9918
surgery1	0.5288	1.8912	0.2574	1.0860
transplant1	0.9898	1.0103	0.5352	1.8307

Concordance= 0.636 (se = 0.033)
 Likelihood ratio test= 15.11 on 4 df, p=0.004
 Wald test = 14.49 on 4 df, p=0.006
 Score (logrank) test = 15.03 on 4 df, p=0.005

Figure 5: Full Cox-PH Model Summary using heart dataset.

Looking at the Schoenfeld residuals for the model plots in figure 9, it was deduced that the variables are not constant over time, and the model was clearly not the correct model to fit.

To figure out the right model, the original study design and the biological interactions between the covariates in the heart dataset were considered.

The following is a quote from the original Crowley and Hu paper:

“Briefly, patients are admitted to the Stanford program (given personal and family consent) after a medical conference at which it is decided that the patient is not likely to respond to other forms of therapy. A donor heart, matched on blood type, is then sought; this search has lasted anywhere from a few days to almost a year. Some patients die before a suitable heart is found. The total number of patients waiting when a heart is found is generally one or two, though it has been as high as seven.”

From this quote, it was deduced that transplant should be considered as a time dependent variable, as each patient upon admission must wait a while for a correct matching donor heart to be found. Additionally, biological interactions were considered. The interaction between year and transplant was chosen to be added to the model, as generally, patients who are admitted first receive the heart

transplant before patients who are placed on the list after them.⁶

The second model, considering the time dependency and the interaction stated above, was clearly a more suitable fit.⁷ Model summary is shown in figure 6.

```
Call:
coxph(formula = Surv(start, stop, event) ~ strata(transplant) +
      age + year + surgery1 + year:transplant1, data = heart)

n= 172, number of events= 75
```

	coef	exp(coef)	se(coef)	z	Pr(> z)	
age	0.03306	1.03362	0.01403	2.356	0.0185	*
year	-0.27612	0.75872	0.10746	-2.570	0.0102	*
surgery1	-0.64568	0.52431	0.37190	-1.736	0.0825	.
year:transplant1	0.22793	1.25600	0.14360	1.587	0.1125	

 signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age	1.0336	0.9675	1.0056	1.0624
year	0.7587	1.3180	0.6146	0.9366
surgery1	0.5243	1.9073	0.2529	1.0868
year:transplant1	1.2560	0.7962	0.9479	1.6643

Concordance= 0.647 (se = 0.034)
 Likelihood ratio test= 18.24 on 4 df, p=0.001
 Wald test = 16.93 on 4 df, p=0.002
 Score (logrank) test = 17.91 on 4 df, p=0.001

Figure 6: Time-Dependent Cox-PH Model Summary using heart dataset.

Looking at the Schoenfeld residuals for the model plots in figure 10, all covariates in the model seemed to be constant over time, therefore satisfying the proportionality assumption. At the 5% significance level, we rejected the null hypotheses of the likelihood ratio, Wald, and Score (log-rank) tests.

5 Discussion and conclusions

For this study, survival data of patients on the Stanford heart transplant program were analyzed. The data can be found in the survival package upon loading data(heart). Original data jasa consisted of variables found in table 2. Additionally, a pre-processed dataset heart was available for usage also. Although it is easier to interpret, it is advised to format the raw jasa data for analysis due to uncertainty in how the heart data was formatted from jasa. Moreover, as the dataset consists of right censored observations, it is important to note that survival time is only known until the end of the study period in the original study. In addition, it should be noted that there are vast discrepancies between the categorical variables due to the structure of the original

⁶This model is similarly constructed to the bone marrow transplant example found in (6).

⁷Fitted using strata(transplant) in R.

study. As seen in plot 8, 16 patients have a history of prior bypass surgery, and 69 patients receive the heart transplant out of 103 patients total. Furthermore, as seen in plot 7, more than 50% of the data consist of patients who are between 45-55 years old. Majority of the age data lie within the 35-55 year old group. Due to the disproportionalities mentioned above, there is evidence of sample selection bias, therefore it is recommended for future analyses to gather more data and normalize the discrepancies between covariates.

To begin our analysis, simple Kaplan-Meier estimates were used to consider the differences in survival time from the categorical variables surgery and transplant. It was shown that patients who had a prior history of bypass surgery had a higher chance of survival compared to patients who have not. Furthermore, it was determined that patients who received a transplant had a higher chance compared to patients who have not. Using a log-rank test, at the 5% significance level, the null hypotheses were rejected for both surgery and transplant groups, and we concluded that there are no differences in the Kaplan-Meier estimates between the groups.

This project also considered two different Cox-proportional hazard models. For both models, the proportionality assumption was checked using Schoenfeld residuals. First a full model was built, without considering any biological meaning. Variables age and year were found to be significant, and it was concluded that the model was limited as it failed to capture any changes of a variable whose value may have been different across time.

The second Cox-proportional hazard model considered the definitions of variables in the original study, as well as biological interpretations to determine the interactions between covariates. The covariate transplant was then fitted as a time-dependent variable, and an interaction between year and transplant was added to the model. Age and year were again considered to be significant. At the 5% significance level, we rejected the null hypotheses of the likelihood ratio, Wald, and Score (log-rank) tests for both Cox-proportional hazard models.

For future studies, it is recommended to compare the results of other plots such as the martingale residual plot and Arjas plot. Additionally, plotting score residuals for non-proportional hazards are always recommended when checking for the proportionality assumption as it has several advantages over the other approaches (6).

References

- [1] “Heart transplant.”
- [2] E. L. Kaplan and P. Meier, “Nonparametric estimation from incomplete observations,” *Journal of the American Statistical Association*, vol. 53, no. 282, p. 457–481, 1958.
- [3] D. R. Cox, “Regression models and life-tables,” *Springer Series in Statistics*, p. 527–541, 1992.
- [4] J. Crowley and M. Hu, “Covariance analysis of heart transplant survival data,” *Journal of the American Statistical Association*, vol. 72, no. 357, p. 27–36, 1977.
- [5] J. P. Klein and M. L. Moeschberger, *Survival analysis: Techniques for censored and truncated data*. Springer, 2003.
- [6] L. Li, “Lecture notes in survival analysis, stat 432/532.,” 2022-2023.
- [7] A. Kassambara, *Drawing Survival Curves using 'ggplot2'*, 2023. R package version 0.4.9.
- [8] T. M. Therneau, *A Package for Survival Analysis in R*, 2023. R package version 3.5-5.

6 Appendix

All code used to create plots and complete analyses can be provided upon request at sp8@ualberta.ca.

6.1 Data

The following table contains the list of definitions in the `java` dataset:

Table 2: Definition of Variables in heart dataset

Variables	Definitions
birth.dt	birth date
accept.dt	acceptance into program
tx.date	transplant date
fu.date	end of followup
fustat	dead or alive
surgery	prior bypass surgery
age	age in years
futime	followup time
wait.time	time before transplant
transplant	transplant indicator (1 = yes)
mismatch	mismatch score
hla.a2	particular type of mismatch
mscore	another mismatch score
reject	rejection occurred

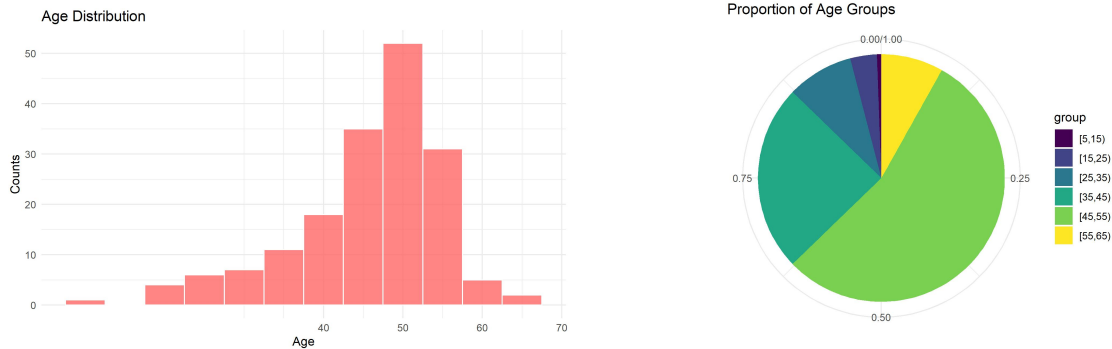


Figure 7: Distribution of age in `java`.

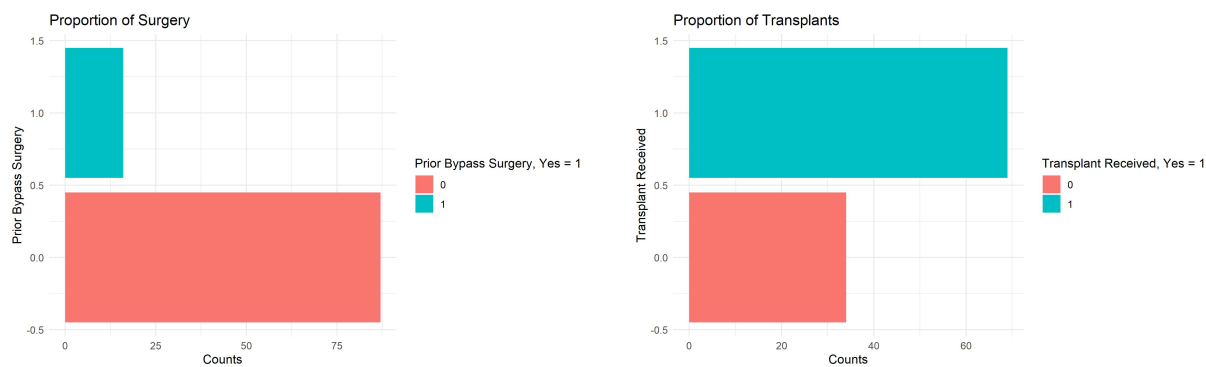


Figure 8: Proportion of factor variables in heart.
Left: Surgery group, Right: Transplant group

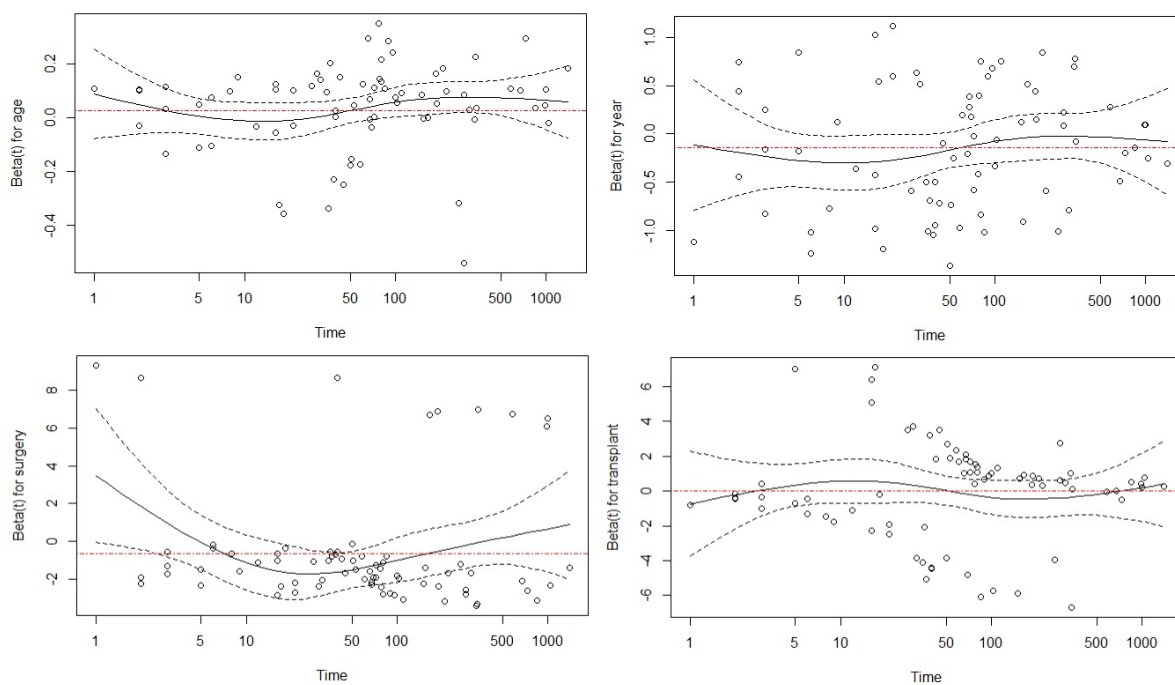


Figure 9: Schoenfeld Residual Plots of Full Cox-PH Model.

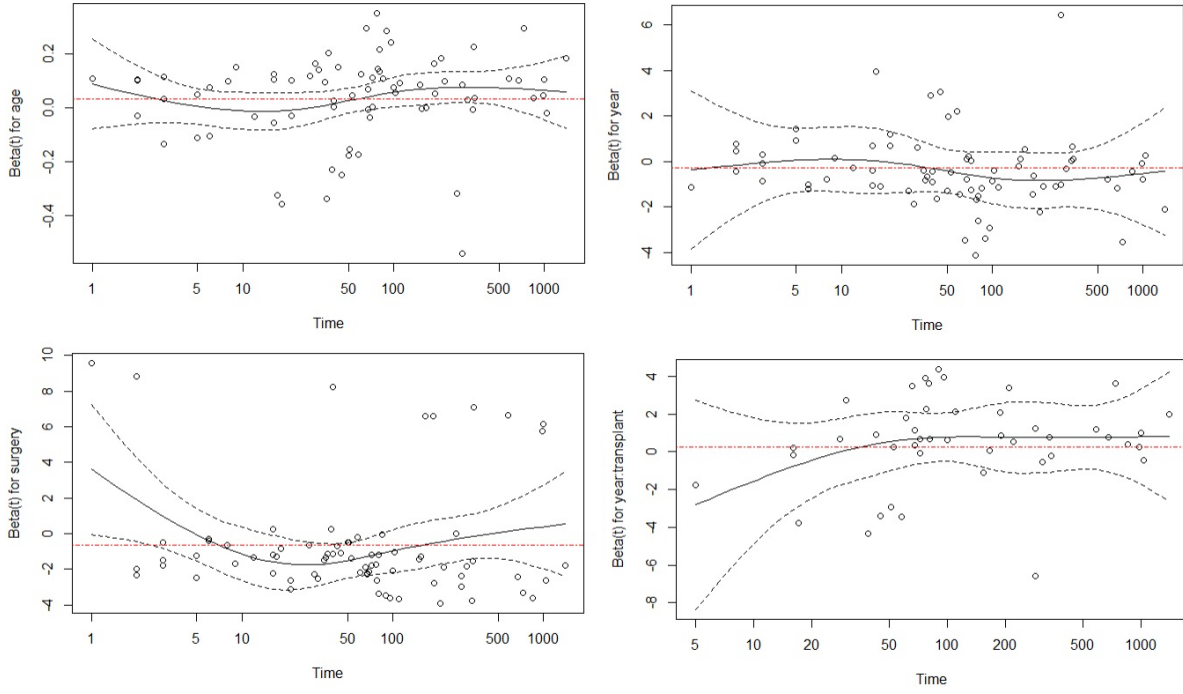


Figure 10: Schoenfeld Residual Plots of Time-Dependent Cox-PH Model.